L21: Power and Inference in Practice

Recap - confidence intervals and testing
α β and types of error
Calculating power in R
Size of the difference
Example From Pagano and Gauvreau "Principles of Biostatistics"
Size of the sample
Calculating sample size
Determining sample size for a margin of error

Recall the simple conditions for inference

1. Simple Random Sample (SRS)
2. Population has a Normal distribution
3. σ is known

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Condition 1: Where do the data come from?

When you use statistical inference, you are acting as if your data are a probability sample or come from a randomized experiment.

Sometimes we treat non-random samples as if they were random. We can do this if we believe that the non-randomness of the sample does not affect the outcome of interest.

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Condition 1: Where do the data come from?

A neurobiologist is interested in how visual perception can be fooled by optical illusions.

A sociologist is interested in attitudes toward the use of human subjects in experiments.

They both use their students as a convenience sample. Can we act as if these students are a SRS? It is easier to make this argument for one of these research questions.

Which one? And why?

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Condition 1: Where do the data come from?

Even if the data come from a randomized experiment, there could be issues that hinder the randomness - we covered some of these when we talked about study design:

- Non response
- Dropout ("Lost to follow-up", in RCT jargon)
- Social desirability

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Condition 2: What is the shape of the population distribution?

- We have leeway around this condition. This is because the z-test (and others we'll learn) rely on the Normality of the sample mean, which is guaranteed by the Central Limit Theorem when n is "large enough"
- The z-test is reasonably accurate for any symmetric distribution if the sample size is moderate
- If the distribution is skewed, we need a large enough n for the z-test to work.

Condition 2: What is the shape of the population distribution?

- Examine the shape of the sample's distribution using a QQ plot or a histogram. Use it to infer the shape of the population
- Difficult to infer much if there are too few observations.

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Condition 2: What is the shape of the population distribution?

- Outliers can affect tests of non-resistant measures, like the mean.
- Double check if the outlier is "real" or an error.

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How confidence intervals behave

Recall the form of a CI:

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$$

Where $z^* \frac{\sigma}{\sqrt{n}}$ is the margin of error.

How confidence intervals behave

The margin of error gets smaller when:

- z* is smaller (i.e., you change to a smaller confidence level). Thus, there is a trade-off between the confidence level and the margin of error.
- σ is smaller. You might be able to reduce σ if there is measurement error. Often times, the σ can't be reduced, it is just a characteristic of the population
- n is larger.

The margin of error only accounts for sampling error

- This is one of the most important points!
- If you're taking epidemiology, you've likely learned about other main errors: confounding, measurement error, and selection bias. These are systematic errors. The confidence interval does not account for systematic errors.

How hypothesis tests behave

- The p-value of a test is dependent on whether H_a is one-sided or two-sided.
- The p-value for a one sided test is one-half the p-value for the two-sided test of the same H_0
- The sample data is used as evidence. It should never be the basis for determining the direction of the alternative hypothesis.

How hypothesis tests behave

- Statistical significance depends on sample size (since sample size determines the standard error of the sample mean)
- Recall the form of the z-test:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{\text{magnitude of observed effect}}{\text{size of chance variation}} = \frac{\textit{signal}}{\textit{noise}}$$

- The numerator quantifies the distance between what you observe in the sample and the null hypothesized parameter.
- The denominator represents the size of chance variations from sample to sample

How hypothesis tests behave

- Statistical significance depends on:
  - The size of the observed effect ($\bar{x} - \mu$)
  - The variability of individuals in the population (σ)
  - The sample size (n)
  - Your criterion for rejecting the null (α)

If you obtain a small p-value it is not necessarily because the effect size is large.

How hypothesis tests behave

Very tiny effects can be deemed statistically significant when you have enough data. This is a big problem in the age of big data, because you're almost guaranteed to obtain statistically significant results, no matter the effect size.

On the other hand, if your sample size is too small, you might not obtain statistical significance even if your effect size is large. Thus "an absence of evidence is not evidence of absence", or failing to reject the null hypothesis does not imply that the null hypothesis is true.

α β and types of error

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α

- The significance level α is the threshold we choose before looking at the data for how much evidence we require before rejecting H_0.
- If the null hypothesis is true, α is the probability that our sample still lands in the rejection region and we reject H_0 anyway.
- Rejecting a true null hypothesis is called a type I error, so α is the type I error rate we are willing to tolerate.
- The p-value is a different thing. It is computed *from the data*. We reject when the p-value is smaller than α.
- But there is another type of error.

Type I error, and Type II error in hypothesis tests

- β is the chance of making a wrong decision when the alternative hypothesis is true.
- This is known as a type II error.

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Type I error, and Type II error in hypothesis tests

	H_a is true	H_0 is true
Reject H_0	Correct decision	Type I error
Fail to reject H_0	Type II error	Correct decision

This table should remind you of something we have seen before....

Screening tests

	Have the disease	Do not have the disease
Test positive	True positive	False positive
Test negative	False negative	True negative

In the case of screening tests, we calculated sensitivity as the P(Test positive | Have the disease)

In the case of hypothesis testing we are often interested in power

Power

- The power is the chance of making the correct decision when the alternative hypothesis is true.
- Thus, it is the complement of β
- Power $= 1 - \beta$

	H_a is true	H_0 is true
Reject H_0	Correct decision	Type I error
Fail to reject H_0	Type II error	Correct decision

Beta and Power

Type II error rate (β) and power are probabilities when H_0 is false

$\beta = \text{P(fail to reject } H_0\text{) when } H_0 \text{ is false}$

$\text{Power} = 1 - \beta = \text{P(reject } H_0\text{) when } H_0 \text{ is false}$

However, there are an infinite number of possible values that μ could assume that are not $= \mu_0$

Thus we must choose a value at which to evaluate the β and power for an alternative hypothesis...

When we evaluate β we do so at a single such value μ_1

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Example of calculating power

Suppose you have a known standard deviation $\sigma = 1$, with $H_0 : \mu = 0$ vs. $H_a : \mu > 0$, and you choose $\alpha = 0.05$. Calculate the power to detect a true mean of $\mu_1 = 0.8$ when $n = 10$.

First find the cutoff on the scale of $\bar{x}$. We reject H_0 for any sample mean in the top 5% of the null sampling distribution, which is centered at 0 with standard error $\sigma/\sqrt{n}$:

```
se <- 1 / sqrt(10)
x_star <- qnorm(p = 0.05, mean = 0, sd = se, lower.tail = FALSE)
x_star
```

```
## [1] 0.5201484
```

Any sample mean at or above this value leads us to reject H_0. On the z scale this same cutoff is $Z_\alpha = 1.645$.

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Example of calculating power

Now suppose that H_a is true and the real mean is 0.8. The test will reject H_0 about what percent of the time? Take the cutoff from the previous slide and calculate the *probability* above it under the alternative sampling distribution:

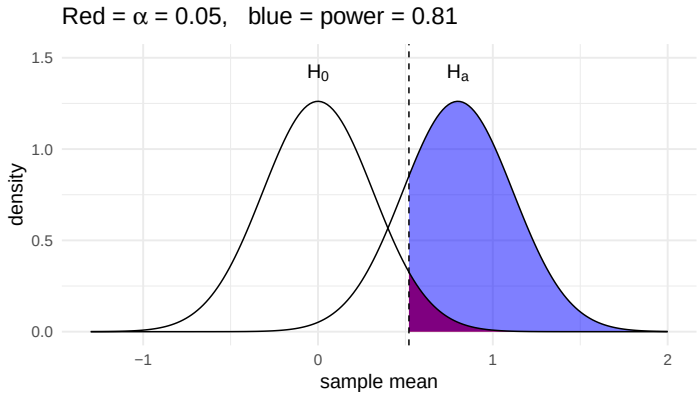
```
pnorm(q = x_star, mean = 0.8, sd = se, lower.tail = FALSE)
```

```
## [1] 0.8119132
```

The power is about 0.81. With $n = 10$ we would detect a true mean of 0.8 about 81% of the time, and we would miss it about 19% of the time. That 19% is β.

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Example of calculating power, illustrated



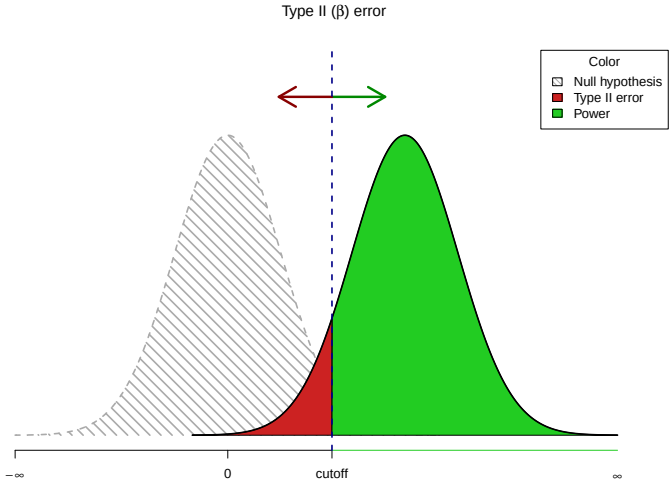
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Effect of changing the μ_1

Imagine our H_0 is a standard normal (mean=0, SD=1) and we set our α at 0.05. If the true mean of our sampled population is 1.7 standard deviations above μ_0, what does our β look like?

Effect of changing the μ_1



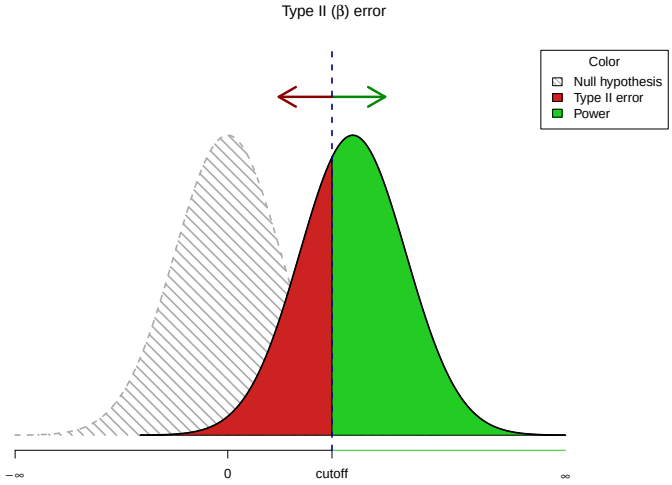
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Effect of changing the μ_1

What happens if the "true" mean is closer to the Null?

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Effect of changing the μ_1

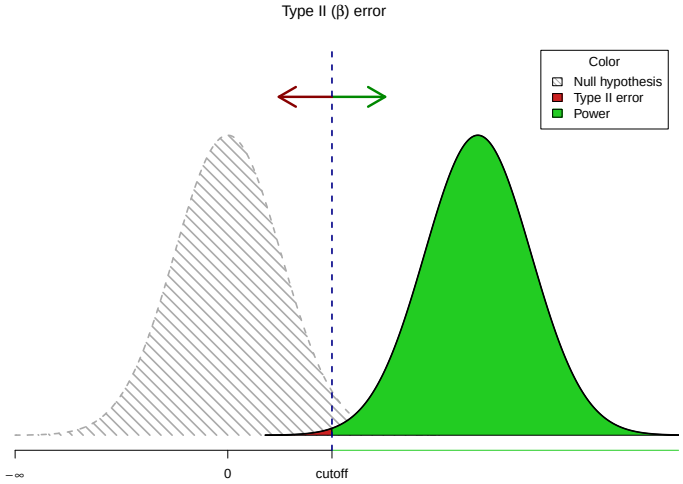


Effect of changing the μ_1

What happens if the “true” mean is further from the null?

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Effect of changing the μ_1



Example From Pagano and Gauvreau "Principles of Biostatistics"

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Example: from Pagano text ch. 10

Suppose we have a sample of 25 males aged 20 to 74 from the United States. We know that 20-24 year old males have a mean serum cholesterol level of 180 mg/100ml. Because cholesterol tends to increase with age, we would expect the mean cholesterol in our sample of 20-74 year olds to be higher than 180 mg/100ml. We also assume the standard deviation in the population is 46 mg/100ml.

$$H_0 : \mu = 180 \qquad \text{vs.} \qquad H_a : \mu > 180$$

Example: from Pagano text ch. 10

This is a one-sided test with $\alpha = 0.05$, so we can use R to get the critical value on the z scale:

```
qnorm(0.95, mean = 0, sd = 1)
```

```
## [1] 1.644854
```

H_0 would be rejected if z test statistic ≥ 1.645

Then at what mean value of cholesterol would we decide to reject the null?

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Example: from Pagano text ch. 10

$$z = \frac{\overline{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

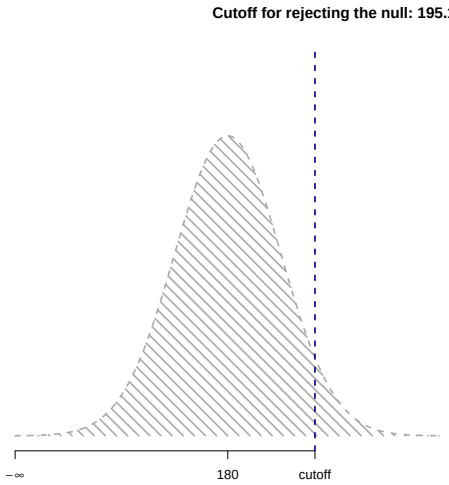
$$1.645 = \frac{\overline{x}^* - 180}{\frac{46}{\sqrt{25}}}$$

Solve this for $\overline{x}^*$

$$\overline{x}^* = 1.645 \times \frac{46}{\sqrt{25}} + 180 = 195.1$$

Example: from Pagano text ch. 10

So here we have our null distribution with the value at which we reject the null



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Example: from Pagano text ch. 10

We must choose a value at which to evaluate β. Here we will choose an alternate hypothesis that $\mu_1 = 211$. Since we know a sample mean less than 195.1 causes us to fail to reject H_0 we need to calculate the proportion of a distribution centered at 211mg/100ml that would be below this value.

$$Z = \frac{195.1 - 211}{\frac{46}{\sqrt{25}}}$$

$$Z = -1.73$$

Example: from Pagano text ch. 10

Using R to calculate the probability,

```
pnorm(-1.73, mean = 0, sd = 1)
```

```
## [1] 0.04181514
```

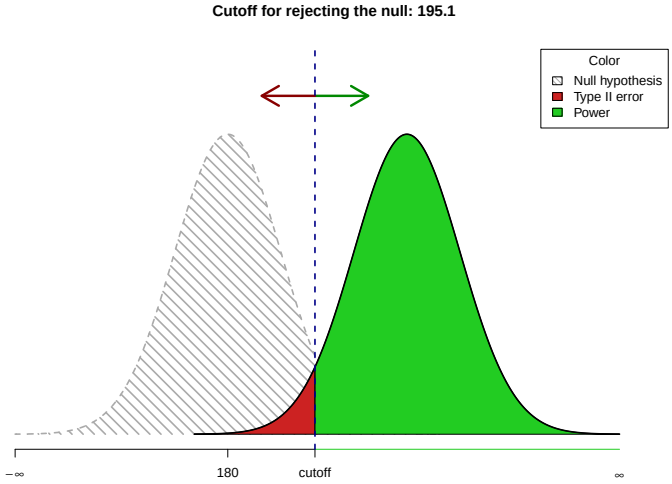
Thus when null is false and true population mean is 211,

$$\beta = P_{\mu=211}(\text{do not reject null (180) }) \approx 0.042$$

Remember that Power is 1-β = $P_{\mu=211}$(reject null)

In this example, Power is 1-0.042 = 0.958

Example: from Pagano text ch. 10



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Size of the difference

So what have we learned about power and β ?

If we increase the size of the difference we are looking for, what happens to power?

What happens to β as we increase the size of the difference?

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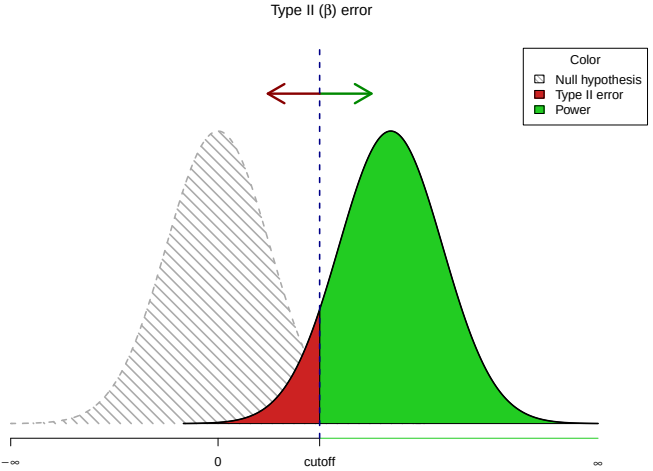
Size of the sample

Remember what happens to our test statistic as sample size increases

$$z = \frac{\overline{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

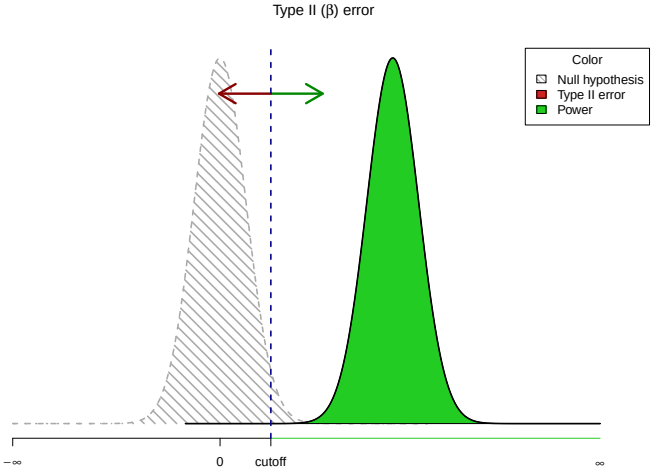
Increasing sample size

If we look at this graph from earlier in the slides, what do we expect to change if we increase our sample size from 25 to 100?



Increasing sample size

Quadrupling n halves the standard error, so both curves get narrower while the distance between their centers stays the same.



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Sample Size

So how does increasing sample size impact power and β?

If we increase sample size we expect β to ?

And we expect power to ?

How might this affect our decisions when we are planning a study?

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Sample Size and phase of clinical trials

From FDA.com

	Size	Length of study	Purpose
I	20-100	Months	Safety and dosage
II	up to several hundred	Months to $\sim$ 2 years	Efficacy and side effects
III	$\sim$300 to 3,000	1-4 years	Efficacy and adverse effects
IV	Thousands being treated	Ongoing	Safety and efficacy

What would you guess here about the size of the difference that would cause you to stop a trial at each of these phases?

How do we use what we have learned about power to plan our studies?

Calculating sample size

Calculating Sample Size

While you may be asked after the fact what power you had to detect a difference in your study, when you are planning a study you would usually want to figure out what sample size would be necessary to provide a specified power for a one sided Z test.

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Sample size when conducting a hypothesis test

To think about sample size for a z-test, four things matter:

- Significance level α : How much protection do we want against getting a statistically significant result from our sample when there really is no effect in the population?
- Effect size: How large an effect in the population is important in practice?
- Power $(1 - \beta)$: How confident do we want to be that our study will detect an effect of the size we think is important? I.e., what is the probability of rejecting H_0 when the alternative hypothesis is true?
- variability in the population: Remember that the underlying variability in our population affects the variability of our sample mean

Example

For example, using our previous study of mean serum cholesterol levels, suppose we assume the following. Note that we are now being stricter than before and using $\alpha = 0.01$ rather than 0.05.

$H_0 : \mu = 180$ mg/100ml

α: 0.01

σ: 46

If the true population mean is as large as 211 and we want to risk only a 5% chance of failing to reject the null, so $\beta = 0.05$ and power would be $= 1 - \beta = 0.95$

Calculating Sample Size

We start by finding the Z value at which we would reject H_0 at $\alpha = 0.01$

We call this value Z_α

```
qnorm(0.01, lower.tail = FALSE)
```

```
## [1] 2.326348
```

Solve for $\overline{x}$

$$2.326 = \frac{\overline{x} - 180}{\frac{46}{\sqrt{n}}}$$

$$\overline{x} = 180 + 2.326\left(\frac{46}{\sqrt{n}}\right)$$

Calculating Sample Size

Now look at the same cutoff from the point of view of the alternative. If the truth is $\mu_1 = 211$ and we are willing to miss it only 5% of the time, the cutoff must sit 1.645 standard errors *below* 211.

We call this value Z_β

```
qnorm(0.05, lower.tail = FALSE)
```

```
## [1] 1.644854
```

Solve for $\overline{x}$

$$-1.645 = \frac{\overline{x} - 211}{\frac{46}{\sqrt{n}}}$$

$$\overline{x} = 211 - 1.645\left(\frac{46}{\sqrt{n}}\right)$$

Calculating Sample Size

Because the value of $\overline{x}$ represents the same cutpoint for each of these, we can set the two equations equal to each other and solve for n.

$$180 + 2.326\left(\frac{46}{\sqrt{n}}\right) = 211 - 1.645\left(\frac{46}{\sqrt{n}}\right)$$

$$(2.326 + 1.645)\left(\frac{46}{\sqrt{n}}\right) = 211 - 180 = 31$$

$$n = \left(\frac{(2.326 + 1.645) \times 46}{211 - 180}\right)^2 = 34.7$$

As we cannot include 0.7 of a person, the convention is to round up. So we would need 35 people in our sample.

Calculating Sample Size

The more general way to express this sample size formula for a one sided Z test is

$$n = \left(\frac{(Z_\alpha + Z_\beta) \times \sigma}{\mu_1 - \mu_0} \right)^2$$

For a two sided Z test this would be

$$n = \left(\frac{(Z_{(\frac{\alpha}{2})} + Z_\beta) \times \sigma}{\mu_1 - \mu_0} \right)^2$$

```
z_alpha <- qnorm(0.01, lower.tail = FALSE)
z_beta  <- qnorm(0.05, lower.tail = FALSE)
ceiling(((z_alpha + z_beta) * 46 / (211 - 180))^2)
```

```
## [1] 35
```

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Selecting an appropriate sample size

How big does the sample size need to be to give a certain size of confidence interval?

Suppose you want your margin of error to equal m. What sample size do you need to obtain a margin of error of m?

You can re-frame the sample size formula for a two sided hypothesis test

$$n = \left(\frac{(Z_{(\frac{\alpha}{2})} + Z_\beta) \times \sigma}{\mu_1 - \mu_0} \right)^2$$

to a margin of error, where m plays the role of the difference we want to detect and Z_β drops out because a confidence interval makes no power promise

$$n = \left(\frac{Z^* \sigma}{m} \right)^2$$

Example of calculating sample size

Body temperature has a known $\sigma = 0.6$ degrees F. We want to estimate the mean body temperature μ for healthy adults within ± 0.05 F with 95% confidence. How many healthy adults must we measure?

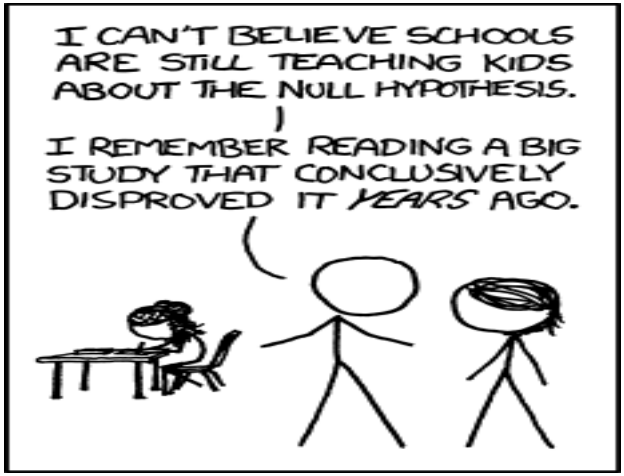
$$n = \left(\frac{z^* \sigma}{m}\right)^2$$

$$n = \left(\frac{1.96 \times 0.6}{0.05}\right)^2 = 553.2$$

We must recruit 554 (round up!) healthy adults for this study.

Parting Humor

From xkcd.com



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